

Response to the Referee

“Formalizing building-up construction of self-dual codes through isotropic lines in Lean”

Jae-Hyun Baek and Jon-Lark Kim

Dear Editor and Referee,

We thank the referee for the careful and mathematically substantive report. The report identified genuine errors in the submitted version, especially the use of monomial equivalence, the characteristic-two hyperbolic-plane claim, and two incomplete independence arguments. We corrected these points in the mathematics, the exposition, and the corresponding Lean statements.

To keep the revision focused, Section 3 retains the original progression from the binary case to the split q -ary case and then to the universal normal forms. The revised hierarchy is explicit: Theorem 3.9 is the adapted one-step reduction and extension, Theorem 3.12 is the forward boxed construction, Theorem 3.13 is its conditional reverse, Theorem 3.15 is the universal rank- r form for every fixed ordered pairing, and Theorem 3.16 proves that a rank-one pairing always exists after a permutation of the individual scalar coordinates. The applications have likewise been limited to a recursive example over $\mathbb{F}_5$ and one complete repeated realization over $\mathbb{F}_{13}$.

The paper-facing Lean development is divided into nineteen independent Challenge/Solution suites containing 181 declarations. The production dependencies use neither `sorry` nor `native_decide`; the transitive axiom audit reports only `propext`, `Quot.sound`, and `Classical.choice`. The finite-field distance enumerations are reproducible exact computations outside Lean and are identified as such in the manuscript.

Below we answer each comment point by point. Referee comments are paraphrased in gray and our responses appear in blue.

Major comments

1. Monomial equivalence and Euclidean self-duality

The full monomial group does not preserve the Euclidean form or Euclidean self-duality. The manuscript must distinguish monomial equivalence from the transformations actually used in the reduction and reconstruction results.

Response. We agree. Definition 2.3 has been replaced by permutation equivalence. The revision proves

$$\langle P_\sigma u, P_\sigma v \rangle = \langle u, v \rangle$$

and therefore the preservation of Euclidean self-duality. It also states explicitly that arbitrary nonzero coordinate scalings need not preserve the Euclidean form or self-duality. Every later occurrence has been audited: the code equivalences in Section 3 are coordinate permutations, while the comparison of two different bilinear spaces in Theorem 3.4 is made by a separately specified form isometry.

The exact Lean interface is permutation-only: `CodeEquiv` is defined using `Equiv.Perm`, and `codeEquiv_preserves_paperSelfDualCode` proves preservation of self-duality.

Location in the revised manuscript: Definition 2.3, p. 5, and the equivalence statements throughout Section 3.

2. The hyperbolic-plane interpretation in characteristic two

The Euclidean plane over $\mathbb{F}_2$ has Gram matrix I_2 and is non-alternating, so it is not isometric to the alternating hyperbolic plane. The cohomological and Euclidean pairings must also be distinguished.

Response. We agree and have removed the erroneous characteristic-two hyperbolic-plane identification. In Theorem 3.4 the Chinburg–Zhang image is first Lagrangian for its cohomological pairing B_{coh} . A specified form isometry

$$\iota_E : (W, B_{\text{coh}}) \longrightarrow (\mathbb{F}_2^{2n}, \langle \cdot, \cdot \rangle_E)$$

then places it in Euclidean coordinates. The deleted coordinate plane has Gram matrix I_2 , is non-alternating, and is explicitly stated not to be isometric to the alternating hyperbolic plane. What the binary calculation uses is the isotropic line $\mathbb{F}_2(1, 1)$, not a hyperbolic pair. The revised proof then establishes the exact deletion and rebuilding identities in these Euclidean coordinates.

The Lean statement `paper_binary_cz_kim_corrected` takes the composite of the form isometry and the selected coordinate permutation as an explicit hypothesis. It returns the Kim condition $\langle x, x \rangle = 1$, the correction coefficients, parent self-duality, and the exact deletion and rebuilding identities. Separate formal lemmas record the two Gram matrices and the characteristic-two non-isometry obstruction.

Location in the revised manuscript: Theorem 3.4, p. 9, together with the abstract, introduction, and conclusion.

3. Definition of Lagrangian subspaces

Maximal total isotropy is not automatically equivalent to $L = L^\perp$ in an arbitrary finite-dimensional bilinear space; the nondegeneracy and dimension hypotheses must be made precise.

Response. We agree. Definition 2.2 now assumes a finite-dimensional vector space with a nondegenerate symmetric bilinear form and defines a Lagrangian by $L = L^\perp$. The valid characterization is stated separately:

$$L = L^\perp \iff L \text{ is totally isotropic and } 2 \dim L = \dim V.$$

Thus even dimensionality is an immediate necessary consequence whenever a Lagrangian exists. The submitted maximal-total-isotropy equivalence has been removed, and “split” is defined separately in the odd-characteristic discussion.

The Lean listing formalizes the orthogonal-complement definition, proves the half-dimension characterization under nondegeneracy, and derives evenness for self-dual codes.

Location in the revised manuscript: Definition 2.2 and the paragraph and Lean listing immediately following it, p. 4.

4. Dimension argument in Theorem 3.9

The submitted proof did not justify independence of the tail vectors, and the expression comparing r_0 with the tail space mixed two ambient spaces.

Response. We agree and replaced that argument. Orthogonality first gives $y_i = \langle x, g_i \rangle$. If $\sum_i \lambda_i g_i = 0$, then

$$\sum_i \lambda_i y_i = \left\langle x, \sum_i \lambda_i g_i \right\rangle = 0,$$

and hence $\sum_i \lambda_i r_i = 0$ in the extended space. Independence of the generator rows therefore gives every $\lambda_i = 0$. Only then do we deduce that the tail space has dimension $n - 1$; its pairwise orthogonality and half dimension imply self-duality. The invalid cross-ambient-space statement has been deleted.

The exact theorem `paper_qary_adapted_reduction` returns the forced coefficients, independence of both row families, tail finrank, tail self-duality, and equality with the building-up family.

Location in the revised manuscript: Theorem 3.9 and its proof, p. 12.

5. Linear independence in Theorem 3.12

The submitted triangular elimination was not proved and might reintroduce previously eliminated entries.

Response. We agree and removed the elimination claim. The revised proof uses the defect functional

$$\delta_c(z) = z_2 - cz_1, \quad \delta_c(t(1, c)) = 0, \quad \delta_c(0, 1) = 1.$$

Applied successively to the first $n - 1$ block columns of a row relation, it reads off all non-final coefficients. The first coordinate of the final block then gives the last coefficient. This proves independence without row elimination.

The Lean map `splitBoxedReadout` implements exactly these readouts. The corresponding exact theorem is

`paper_split_boxed_form_exact.`

It proves pairwise orthogonality, linear independence, and self-duality from the three displayed relations.

Location in the revised manuscript: Theorem 3.12 and its proof, pp. 15–16.

6. Scope of the split q -ary boxed form

The manuscript must distinguish the forward construction, reconstruction for adapted families, a conditional reverse theorem, and a universal normal-form claim, which was not proved in the submitted version.

Response. We agree with the assessment of the submitted version. The revision now separates these five levels rather than describing the adapted result as universal:

- Theorem 3.9 gives the adapted one-step reduction and extension;
- Theorem 3.12 gives the forward split boxed construction;
- Theorem 3.13 gives the reverse implication under an explicit pivot-adapted presentation;
- Theorem 3.15 gives a universal rank- r boxed normal form for every fixed ordered pairing, where $r = \dim(C \cap U_c)$; and
- Theorem 3.16 proves that every nonzero-length Euclidean self-dual code admits a pairing of rank one after a permutation of the $2n$ scalar coordinates.

For a fixed pairing, Theorem 3.15 sets $U_c = \bigoplus_j \mathbb{F}_q(1, c)$ and $r = \dim(C \cap U_c)$, and produces the displayed rank- r matrix with a nonsingular terminal matrix D . Theorem 3.16 then makes the stronger existential assertion

for every $C \neq 0$ there is a scalar-coordinate permutation π such that $\dim(\pi(C) \cap U_c) = 1$.

Thus every nonzero-length Euclidean self-dual code admits a pairing of rank one. This is distinct from the generally false assertion that every preassigned pairing has rank one. The fixed-pairing theorem is obtained from the defect map by rank-nullity and a nonsingular coordinate minor; its converse follows from the displayed Gram relations and an independence readout. The rank-one existence theorem then changes the pairing by the Kim–Lee/direct-sum induction, after which the same Gram calculation gives the displayed rank-one generator and its converse.

The two formal interfaces are

```
every_qary_selfDualCode_has_rankBoxed_normalForm,
every_qary_selfDualCode_has_rankOne_orientedPairing_exact.
```

They require, respectively, the exact fixed-pairing rank, literal boxed row-space equality, and the scalar-coordinate permutation producing rank one; neither universality statement is supplied as an axiom.

Location in the revised manuscript: Theorems 3.9, 3.12, 3.13, 3.15, and 3.16, especially Section 3.3 beginning on p. 19.

Minor comments

1. Proposition 2.2(iv): signs and division by two

The factor order and sign convention should agree with Lean, and the explicit representation of every field element requires that 2 be invertible.

Response. The proposition now assumes q odd and uses consistently

$$(x - cy)(x + cy) = x^2 + y^2.$$

For $a \in \mathbb{F}_q$, it takes $x = (1 + a)/2$ and $y = c(1 - a)/2$, so that $x - cy = 1$, $x + cy = a$, and $x^2 + y^2 = a$. This is the same convention as the Lean theorem `paper_split_consequences_exact`, whose assumptions include $2 \neq 0$.

Location in the revised manuscript: Proposition 2.2, p. 6.

2. Proposition 2.2(i): characteristic two

The assertion that c has order four requires exclusion of characteristic two in the proposition itself.

Response. The proposition now begins “Let q be odd.” Hence $-1 \neq 1$, while $c^2 = -1$ gives $c^4 = 1$, and the proof concludes that c has order four. The same condition appears explicitly in the Lean assumptions.

Location in the revised manuscript: Proposition 2.2, p. 6.

3. Terminology “Euclidean”

The manuscript should explain that “Euclidean” denotes the coding-theoretic dot product and not positive definiteness.

Response. This convention is now stated immediately after the dot product is defined. The cohomological pairing, Euclidean dot product, and alternating hyperbolic form are given separate names whenever more than one is present.

Location in the revised manuscript: Beginning of Section 2, p. 3, and Theorem 3.4.

4. Typographical error in the submitted Theorem 3.6

The rows of the generator matrix should be $g_1, \dots, g_n$.

Response. Corrected. Revised Theorem 3.6 states that the rows are $g_1, \dots, g_n$.

Location in the revised manuscript: Theorem 3.6, p. 11.

5. Equivalence in the converse theorem

The equivalence in the original converse should be checked, because a broader monomial equivalence would not preserve Euclidean self-duality.

Response. We checked Kim–Lee, Proposition 2.2, directly. Its statement is explicitly “up to permutation equivalence.” Revised Theorem 3.8 reproduces that wording, consistently with Definition 2.3.

Location in the revised manuscript: Theorem 3.8, p. 12, and Definition 2.3.

6. The ambient decomposition in Theorem 3.9

The first two standard coordinate vectors are not the hyperbolic basis from Proposition 2.3. The proof must say whether the decomposition is abstract or changes the displayed matrix.

Response. The revised statement distinguishes these points. The displayed coordinates give the literal orthogonal decomposition $W = E \perp W_0$, where the standard basis of E has Gram matrix I_2 . Because q is odd and $c^2 = -1$, the vectors $(1, c)$ and $\frac{1}{2}(1, -c)$ give an abstract hyperbolic basis of E . No coordinate change is applied to the displayed building-up matrix.

Location in the revised manuscript: Theorem 3.9(1) and the opening paragraph of its proof, p. 12.

7. Off-diagonal contribution in Theorem 3.12

The proof should display why the two relevant blocks contribute $c(b_{ij} + b_{ji})$.

Response. The revised proof now writes the two products explicitly:

$$(0, 1) \cdot b_{ji}(1, c) = cb_{ji}, \quad b_{ij}(1, c) \cdot (0, 1) = cb_{ij}.$$

It also states that every other non-final contribution is a product of two multiples of the isotropic vector $(1, c)$ and therefore vanishes.

Location in the revised manuscript: Proof of Theorem 3.12, p. 15.

8. Arithmetic discussion in the introduction

The introduction's compressed discussion of étale and compactly supported cohomology should either be explained clearly or reduced to the precise input from Chinburg–Zhang.

Response. We chose the second option. The long exact sequence and the compressed sheaf-theoretic setup have been removed. The introduction now states only the input used later: the global-to-local image is Lagrangian for the sum of local Hilbert pairings, and Chinburg–Zhang's Theorem 1.5 realizes every binary self-dual code of the stated length up to permutation equivalence. The necessary spaces and the form isometry are then defined immediately before Theorem 3.4.

Location in the revised manuscript: Introduction, pp. 1–3, and the paragraph preceding Theorem 3.4.

9. Choice of the set S

The Golay-code example based on a particular set of primes was not connected to the subsequent formalization.

Response. That example and the unexplained list of primes have been removed. The revision mentions S only to state the arithmetic realization input needed for the binary comparison.

Location in the revised manuscript: Introduction and the setup preceding Theorem 3.4.

10. Literature and originality

The manuscript should distinguish the known binary and q -ary building-up theorems from the new interpretation, boxed construction, and formal integration.

Response. The revised text labels Kim’s and Kim–Lee’s results as cited theorems before stating our results. The contribution list now distinguishes the corrected binary comparison through a form isometry, the adapted split reduction, the forward and conditional reverse boxed statements, the universal fixed-pairing rank- r form, the universal rank-one form after coordinate permutation, the selected applications, and the Lean verification. It also states that the arithmetic realization and the cited coding-theoretic theorems remain mathematical inputs rather than formalized new results.

Location in the revised manuscript: Final part of Section 1, especially the contribution list; Section 3.

11. Language and presentation

The manuscript requires proofreading for grammar, notation, and consistency between minimum weight and minimum distance.

Response. We proofread the manuscript and standardized its theorem terminology and the coding-theoretic use of minimum distance. General statements use $\mathbb{F}_q$, whereas $\text{GF}(q)$ is reserved for explicit computational examples. We also state once that the minimum distance of a linear code equals the minimum Hamming weight of its nonzero codewords. The quoted awkward opening sentence and related formulations have been rewritten. During this revision we also corrected the norm hypothesis in the cited q -ary building-up theorem from $x \cdot x = 1$ to $x \cdot x = -1$, in agreement with Kim–Lee, Proposition 2.1 and with the displayed Gram calculation.

Location in the revised manuscript: Section 1; beginning of Section 2; Theorem 3.6; and the manuscript throughout.

We again thank the referee for identifying the mathematical issues that led to these corrections. We believe the revised manuscript now states the scope of each result precisely and gives complete paper proofs that match the formal interfaces.

Sincerely,
Jae-Hyun Baek and Jon-Lark Kim